VisTaxa: Developing a Taxonomy of Historical Visualizations









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Background: Historical Visualizations

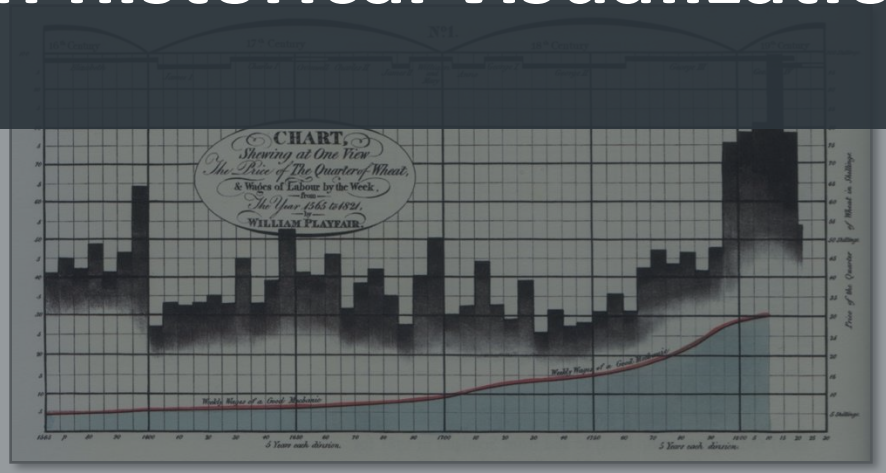


Snow's cholera map, 1855

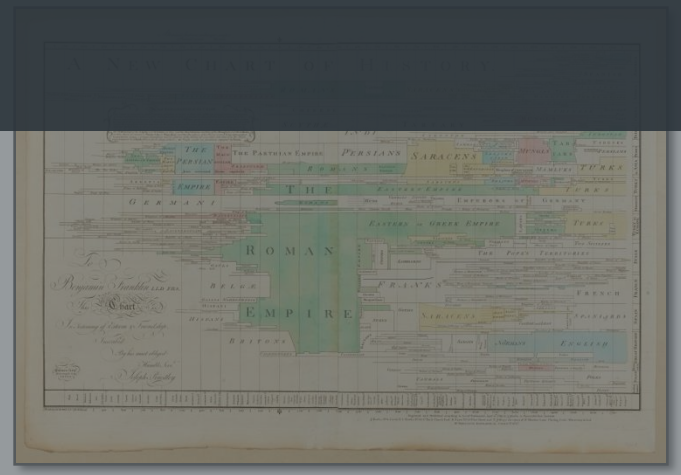


Minard's map of Napoleon's Russian campaign, 1869

What visual designs have been used in historical visualizations?



Playfair's chart of wheat price and wage, 1822



Priestley's chart of history, 1769

Background: Historical Visualization Dataset

- Data-driven approach
- OldVisOnline dataset [1]
  - 13K visualizations before 1950

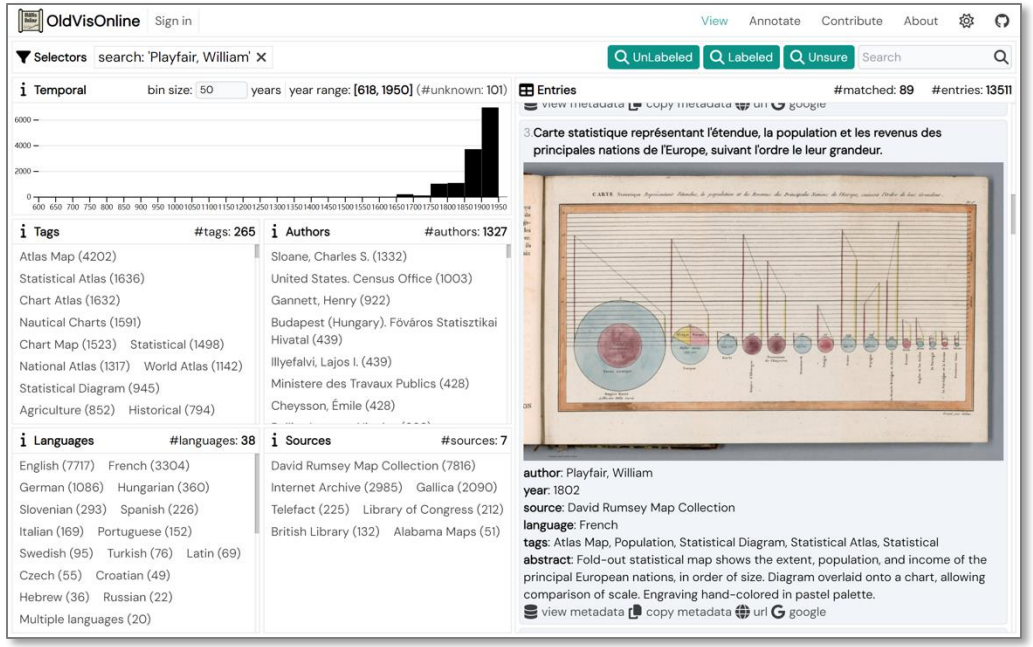
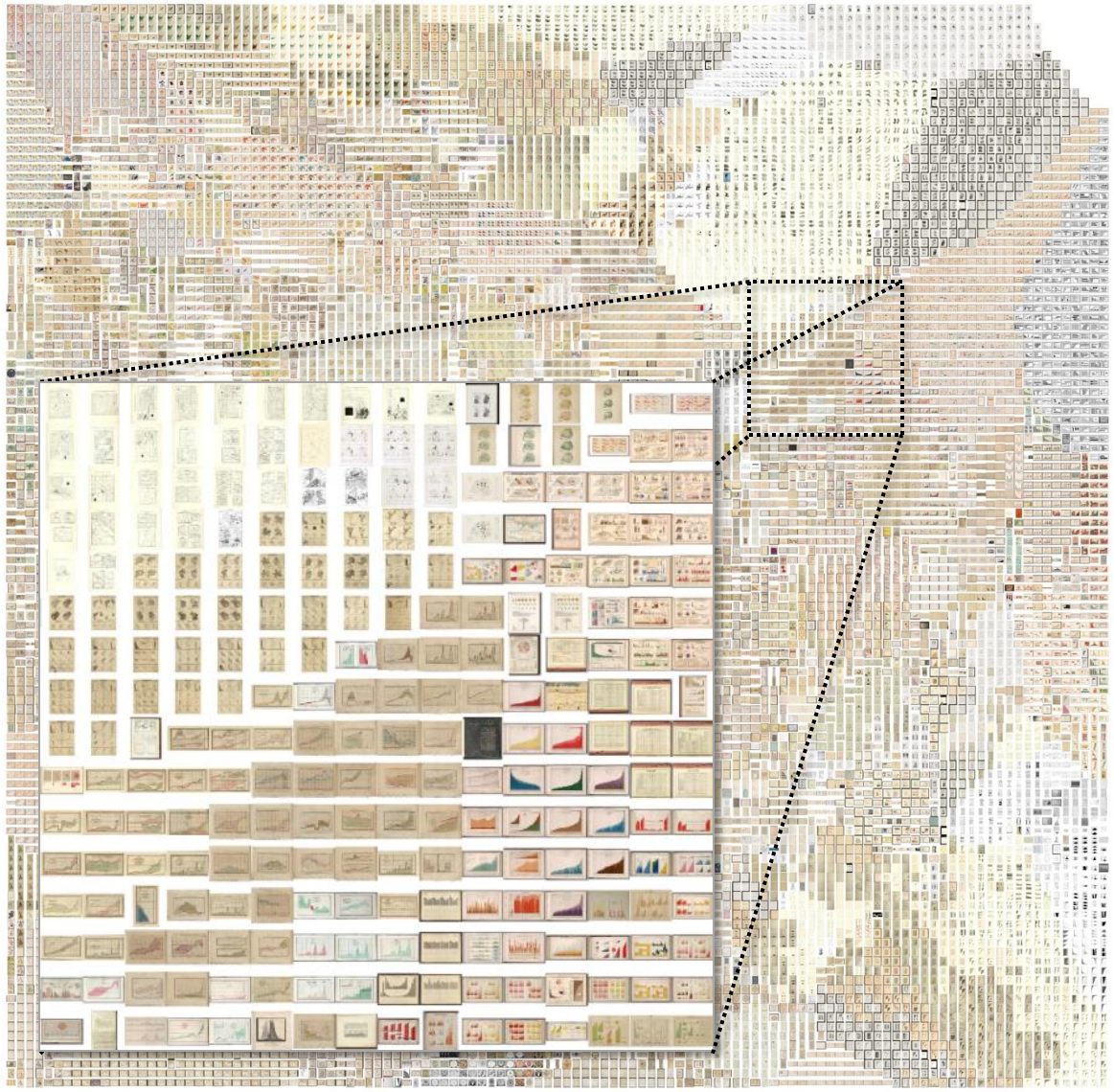


Image Gallery of OldVisOnline



[1] **OldVisOnline: Curating a Dataset of Historical Visualizations**, Yu Zhang et al., IEEE TVCG, 2023

Background: Visualization Taxonomy

- Taxonomy as a tool to organize design space
- Example: Shneiderman's task by data type taxonomy [1]
  - Data types: 1D, 2D, 3D, temporal, multi-dimensional, tree, network
  - Tasks: overview, zoom, filter, details-on-demand, relate, history, extract

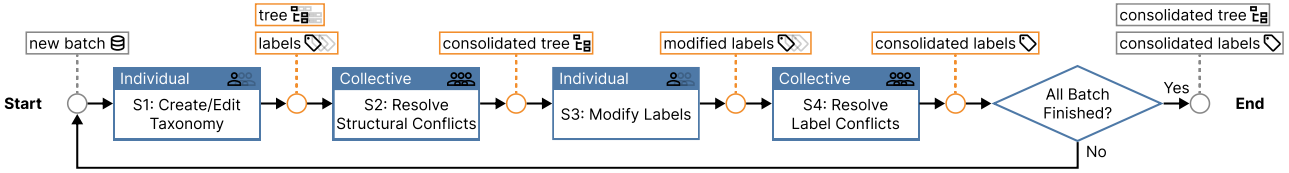
Gap 1: Taxonomy for historical visualizations

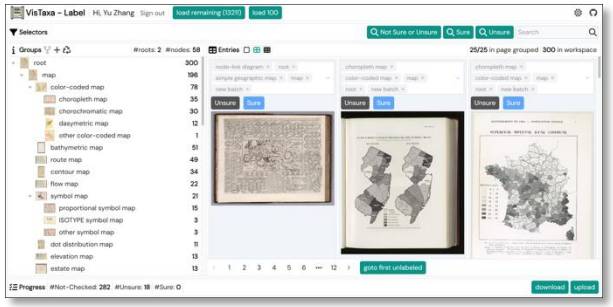
Gap 2: Operationalize taxonomy development

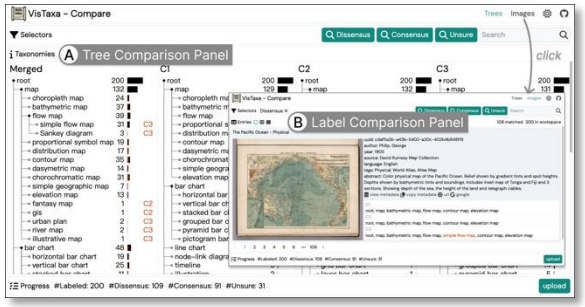
[1] **The Eyes Have It: A Task by Data Type Taxonomy for Information**, Ben Shneiderman, IEEE Symposium on Visual Languages, 1996

Outline

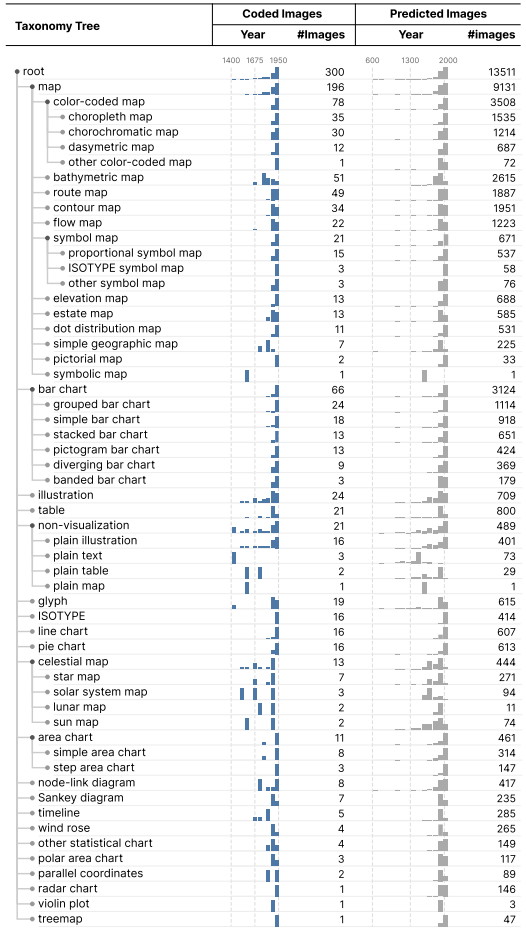
- Protocol and system







- Taxonomy and analysis

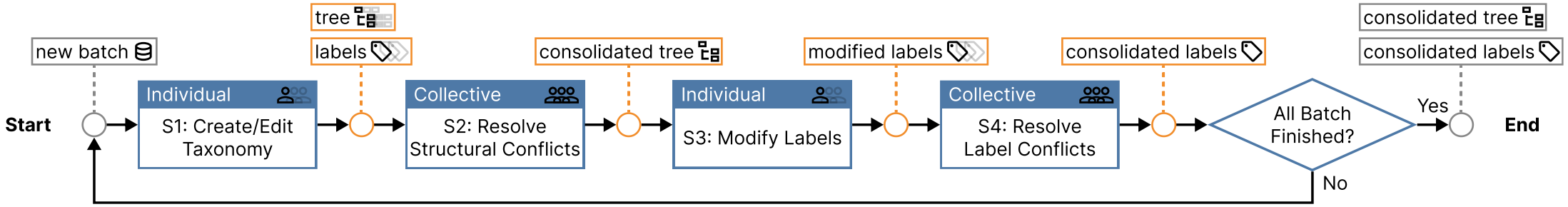


The Taxonomy Development Task

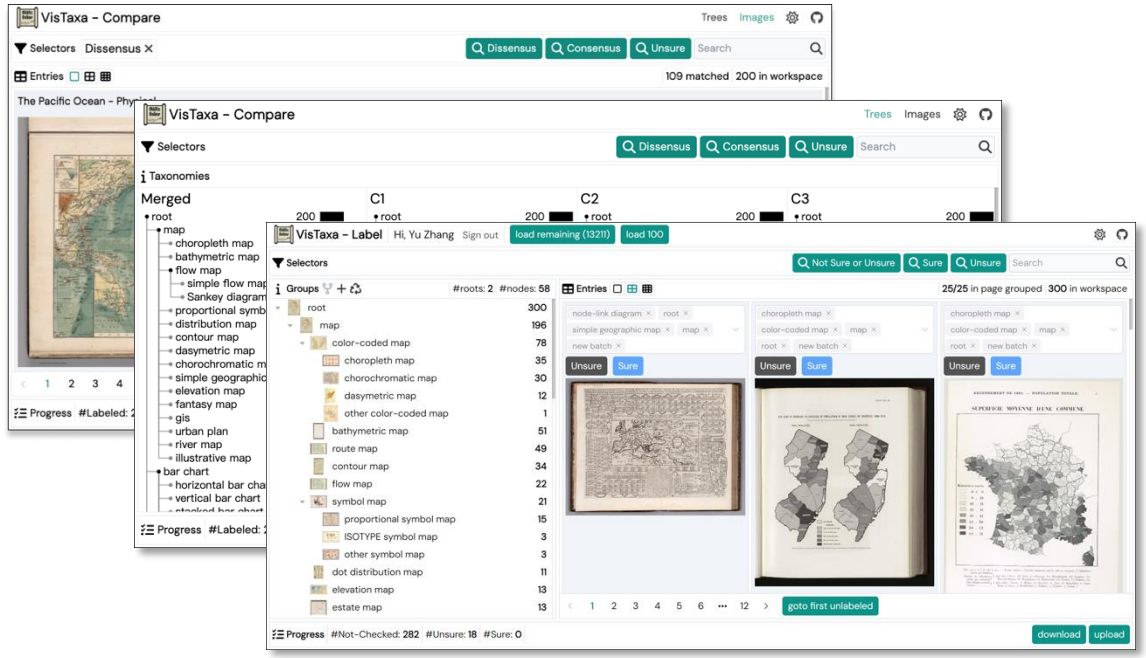
- Construct a hierarchy
  - Build a tree
  - Each node is a taxon
- Assign images to categories
  - Image labeling
  - Assign image to ≥ 1 leaf nodes

Scope: Categorization by visual similarity

A Protocol and the VisTaxa System

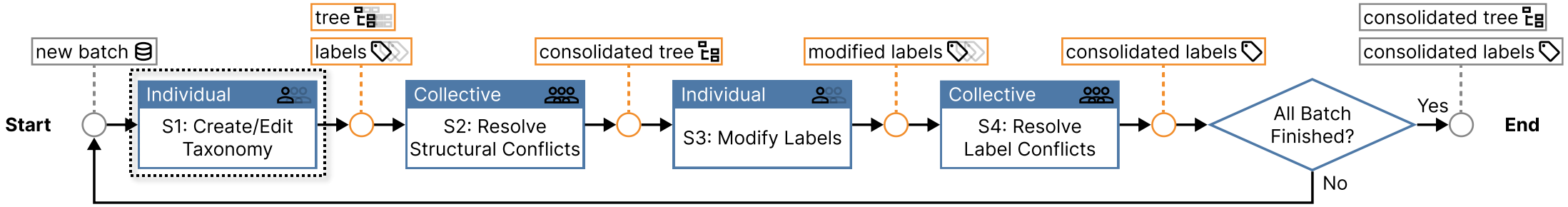


1. Create and edit taxonomy
2. Resolve structural conflicts
3. Modify labels
4. Resolve label conflicts



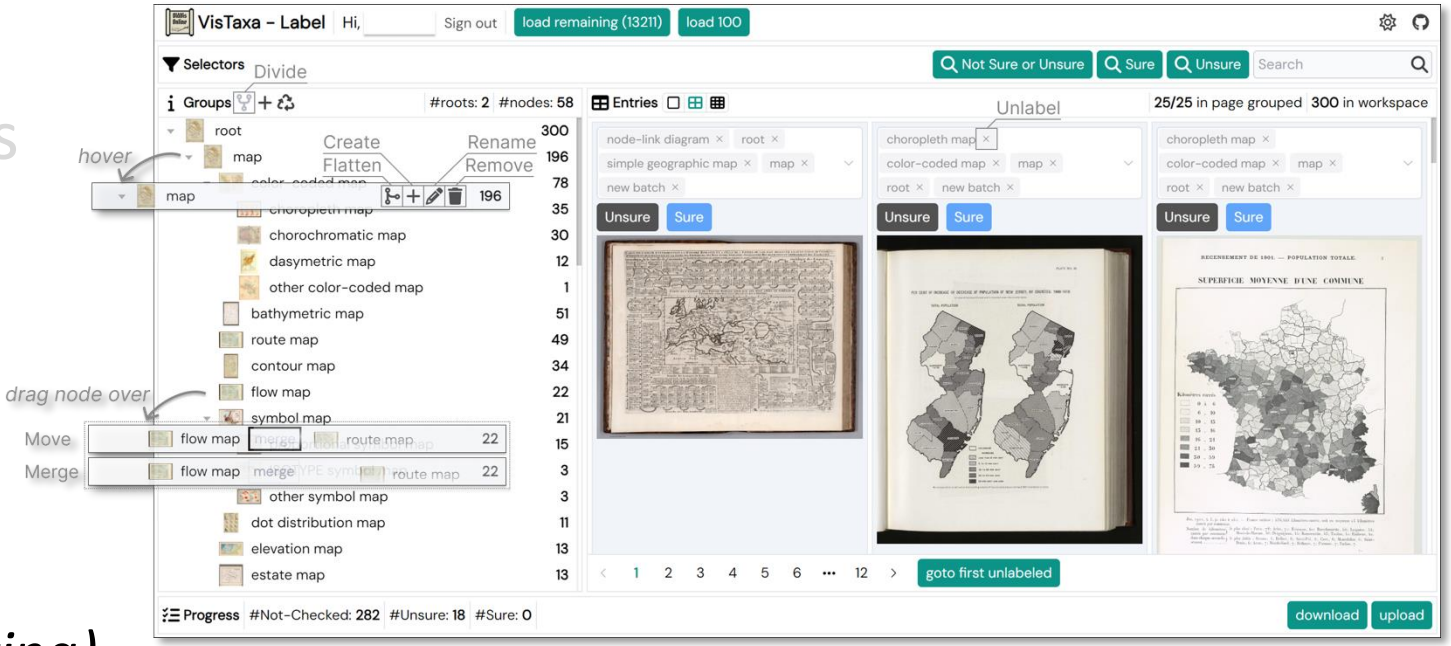
VisTaxa interfaces

A Protocol and the VisTaxa System



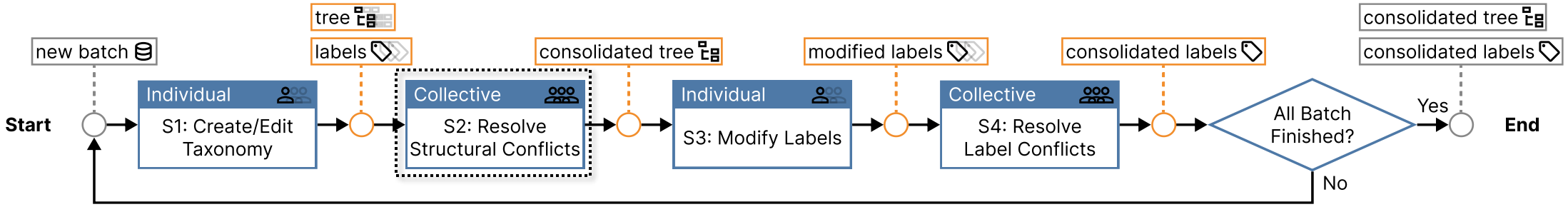
1. Create and edit taxonomy
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Tree editing and label editing with machine assistance (e.g., image clustering)



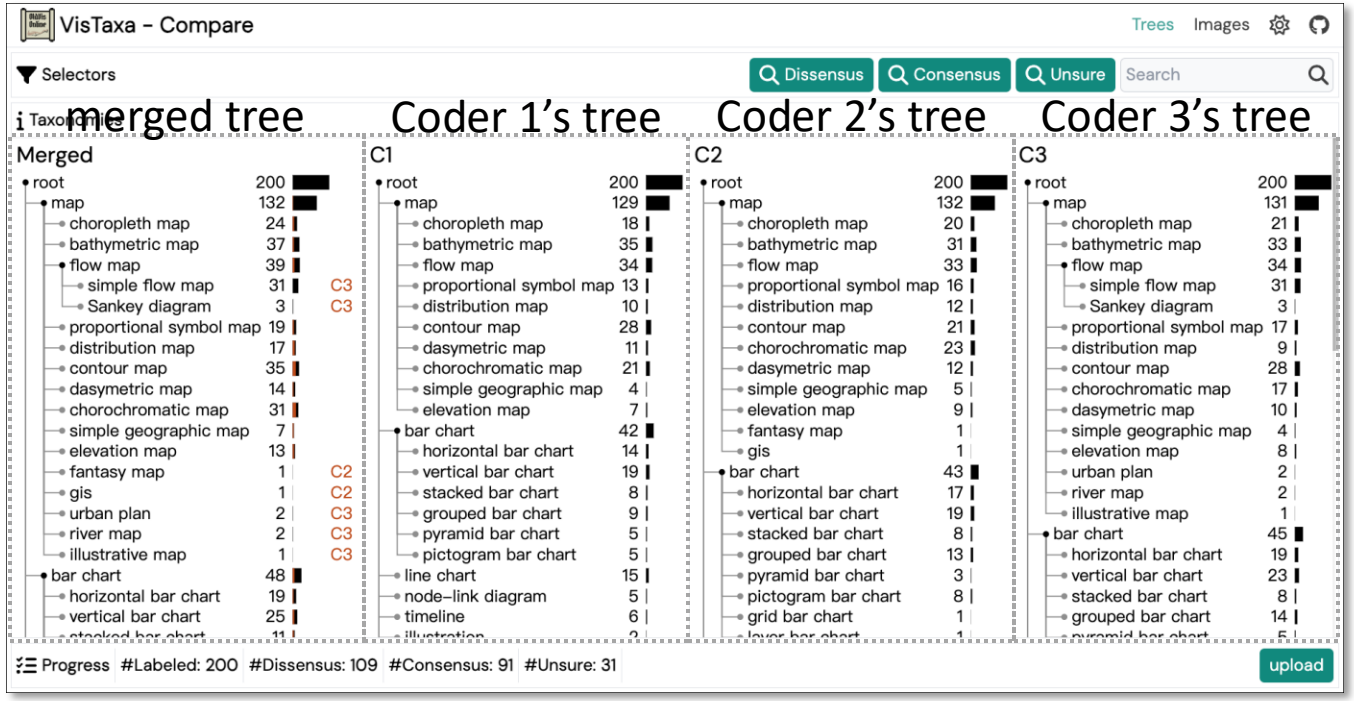
taxonomy labeling interface of VisTaxa

A Protocol and the VisTaxa System



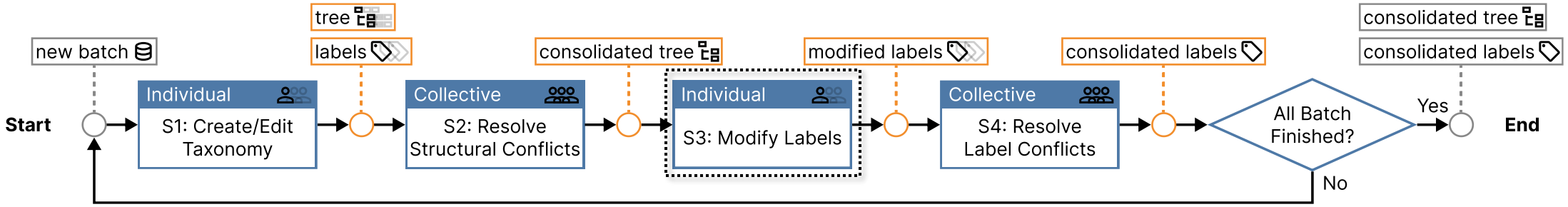
1. Create and edit taxonomy
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Tree comparison



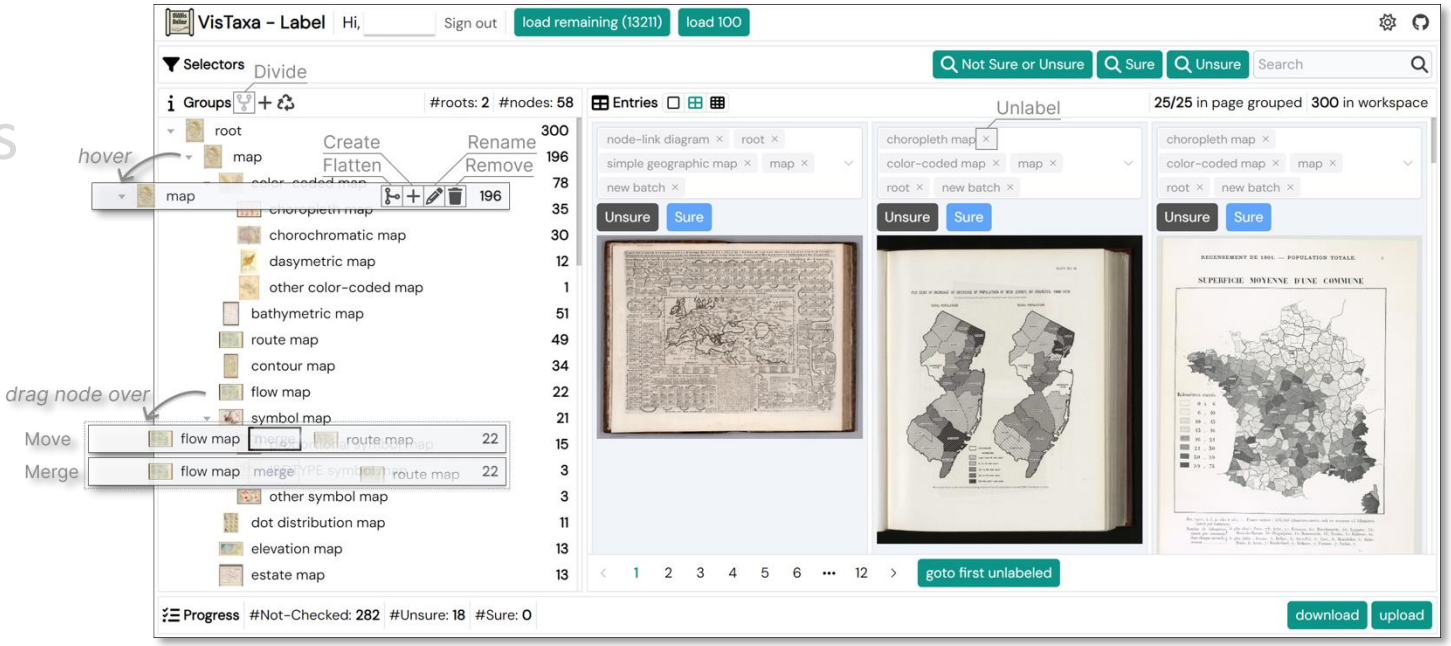
taxonomy comparison interface of VisTaxa

A Protocol and the VisTaxa System



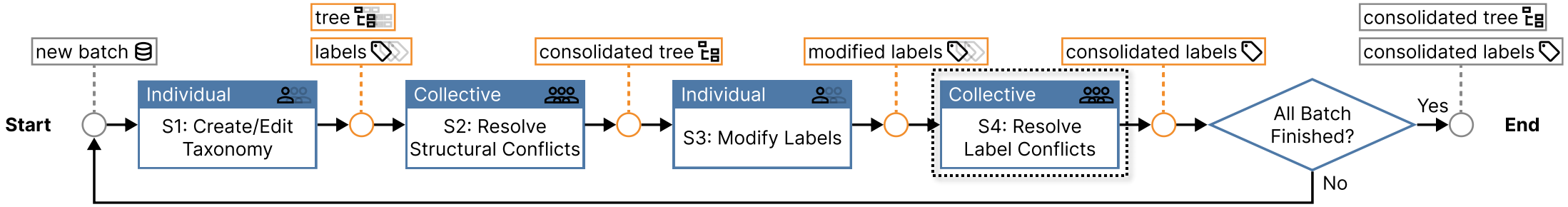
1. Create and edit taxonomy
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label editing



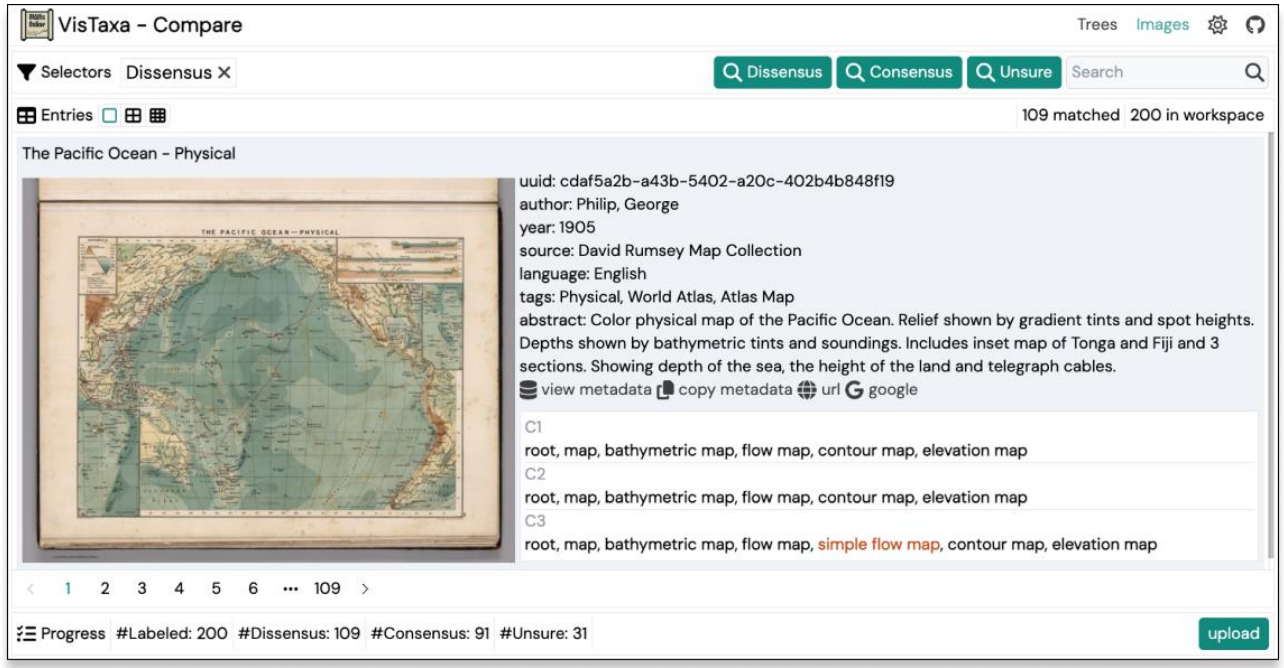
taxonomy labeling interface of VisTaxa

A Protocol and the VisTaxa System



1. Create and edit taxonomy
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label comparison

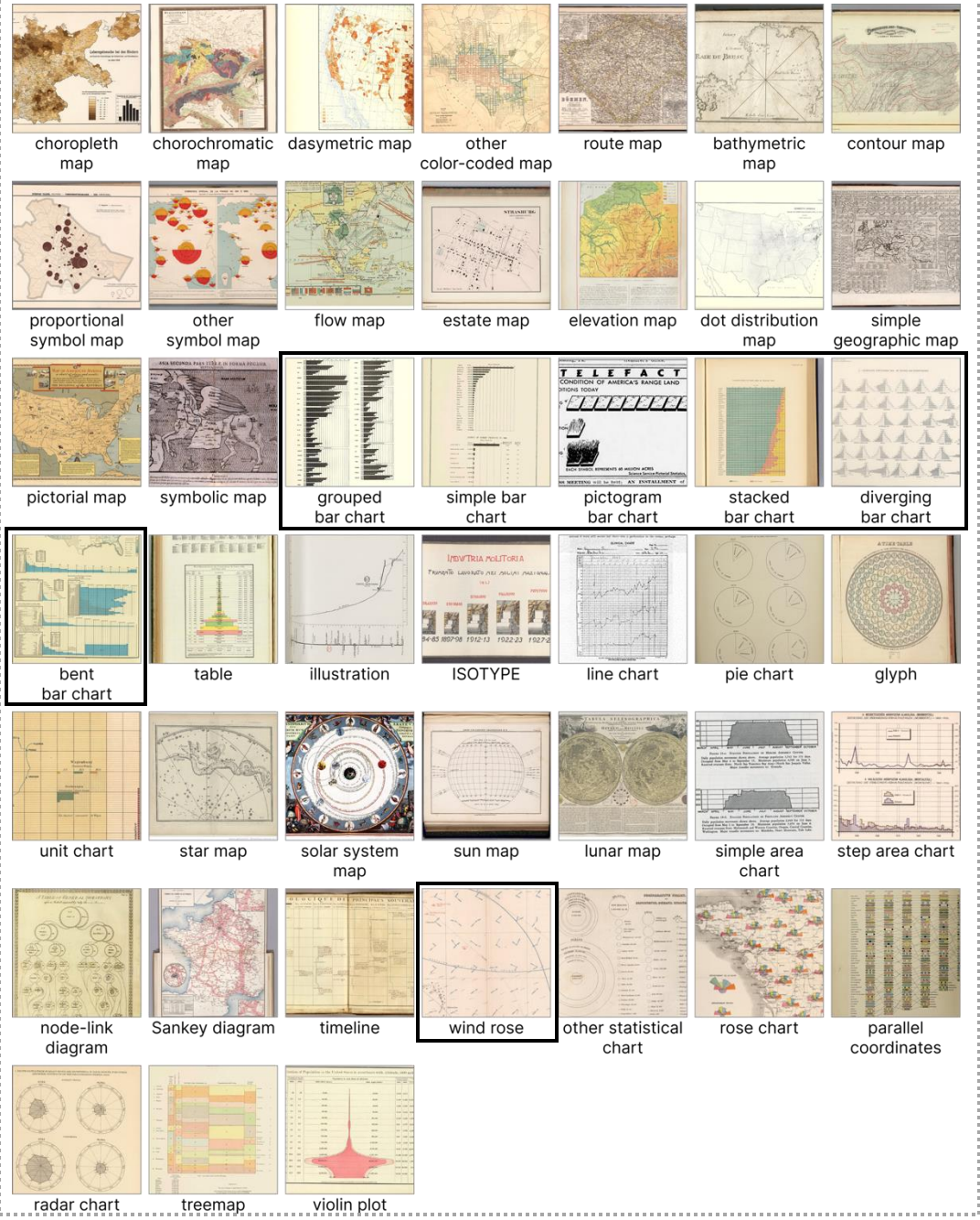


taxonomy comparison interface of VisTaxa

The Resulting Taxonomy

- 3 coders
- 400 images
- **Result:** 51 taxa
  - 45 leaf-level taxa

Taxonomy Tree	Coded Images: Year (550 1250 1950)	#Images
root		400
map		279
color-coded map		121
choropleth map		56
chorochromatic map		45
dasymetric map		18
other color-coded map		2
route map		69
bathymetric map		65
contour map		43
symbol map		30
proportional symbol map		27
other symbol map		3
flow map		27
estate map		24
elevation map		18
dot distribution map		12
simple geographic map		10
pictorial map		2
symbolic map		1
bar chart		88
grouped bar chart		27
simple bar chart		26
pictogram bar chart		24
stacked bar chart		17
diverging bar chart		10
bent bar chart		5
table		31
illustration		28
non-visualization		26
plain illustration		18
plain text		4
plain map		3
plain table		2
ISOTYPE		22
line chart		21
pie chart		21
glyph		19
unit chart		19
celestial map		18
star map		11
solar system map		4
sun map		2
lunar map		2
area chart		11
simple area chart		8
step area chart		3
node-link diagram		11
Sankey diagram		9
timeline		7
wind rose		4
other statistical chart		4
rose chart		3
parallel coordinates		2
radar chart		1
treemap		1
violin plot		1



Scaling to 13K Images in OldVisOnline

Taxonomy Tree	Coded Images		Predicted Images	
	Year	#Images	Year	#Images
	550 1250 1950		550 1250 1950	
• root		400		13511
• map		279		9360
• color-coded map		121		3787
• choropleth map		56		1632
• chorochromatic map		45		1380
• dasymetric map		18		730
• other color-coded map		2		45
• route map		69		1979
• bathymetric map		65		2498
• contour map		43		1709
• symbol map		30		656
• proportional symbol map		27		619
• other symbol map		3		37
• flow map		27		1066
• estate map		24		766
• elevation map		18		644
• dot distribution map		12		505
• simple geographic map		10		452
• pictorial map		2		30
• symbolic map		1		1
• bar chart		88		3329
• grouped bar chart		27		1005
• simple bar chart		26		986
• pictogram bar chart		24		642
• stacked bar chart		17		659
• diverging bar chart		10		329
• bent bar chart		5		270
• table		31		1056
• illustration		28		652
• non-visualization		26		504
• plain illustration		18		327
• plain text		4		115
• plain map		3		50
• plain table		2		22
• ISOTYPE		22		446
• line chart		21		681
• pie chart		21		633
• glyph		19		540
• unit chart		19		466
• celestial map		18		436
• star map		11		262
• solar system map		4		129
• sun map		2		44
• lunar map		2		7
• area chart		11		395
• simple area chart		8		281
• step area chart		3		114
• node-link diagram		11		353
• Sankey diagram		9		233
• timeline		7		286
• wind rose		4		246
• other statistical chart		4		102
• rose chart		3		78
• parallel coordinates		2		73
• radar chart		1		109
• treemap		1		44
• violin plot		1		3

- Similarity-based matching
  - With CLIP embedding
- Performance

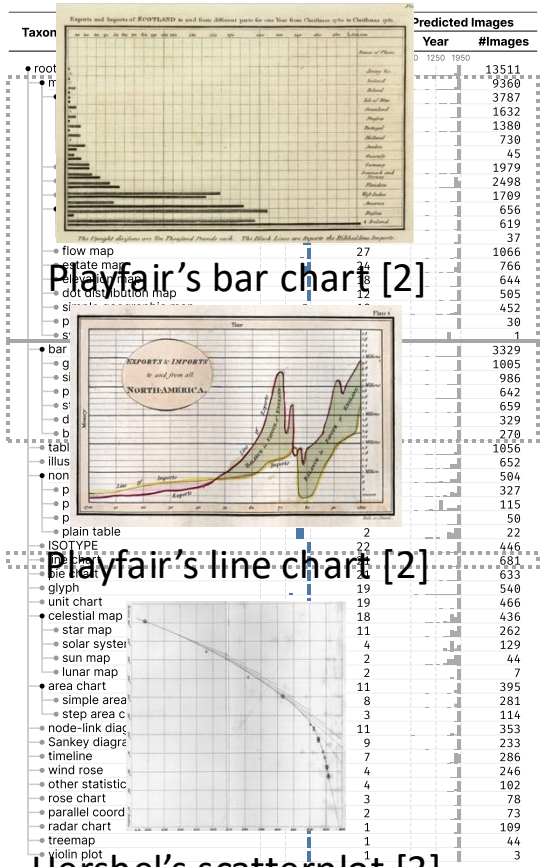
Exact Match	Jaccard Score
39.75%	0.619

Analysis: Frequent Representation Types

- **Contemporary:** bar chart, line chart, scatterplot, map [1]
- **Historical (> 5%):** map, bar chart, table, illustration, ISOTYPE, line chart, pie chart
- Low presence of scatterplots in historical visualization
- Interpretation: late invention of scatterplot
  - Invention of bar chart and line chart: William Playfair, 1786 [2]
  - Invention of scatterplot: John Herschel, 1833 [3]
  - Abstract and statistical thinking [4]

Collection	Size	Total Types	Visualization Type Labels
D3	1247	22	area (32), bar (154), box (11), bubble (70), chord (34), donut (31), heatmap (32), **geographic map (379)**, graph (60), hexabin (21), line (137), radial (15), pie (7), sankey (11), scatter (118), treemap (10),voronoi (25),waffle (12),word cloud (6), sunburst (28), stream graph (13), parallel coordinates (23)
Chartblocks	22750	4	pie (3514), **line (8063)**, bar (7402), scatter (1749)
Fusion Charts	530	10	area (14), **bar (224)**, box (22), donut (54), geographic map (48), heatmap (12), line (84), pie (146), scatter (20), sunburst (0)
Graphiq	2727	11	bubble (9), donut (18), area (210), graph (5), geographic map (244), line (655), waffle (0), box (6), bar (1542), treemap (86), scatter (28)
Plotly	6544	11	area (10), bar (1364), box (259), contour (118), donut (193), filled-line (126), geographic map (184), line (1198), pie (26), radial (17), **scatter (3049)**

Battle et al.'s categorization of chart types in visualization library galleries



[1] **Beagle: Automated Extraction and Interpretation of Visualizations from the Web**, Leilani Battle et al., ACM CHI, 2018
[2] **The Commercial and Political Atlas**, William Playfair, 1786
[3] **On the Investigation of the Orbits of revolving Double Stars**, John Herschel, Memoirs of the Royal Astronomical Society, 1833
[4] **The Early Origins and Development of the Scatterplot**, Micheal Friendly and Daniel Denis, Journal of the History of the Behavioral Sciences, 2005

Limitation and Future Work

- Conclusiveness of the analysis
  - As a source of hypotheses
  - Corpus and machine prediction
- Alternative classification schemes
  - Granularity of the scheme
- Taxonomies (classification schemes) as interfaces
  - Usage scenarios of taxonomies
  - Usability issues of taxonomies

Summary

- A protocol and system
- A taxonomy
- Analysis and reflection

More details on the coding process and result analysis in our paper 👀



Code:
github.com/oldvis/image-taxonomy-labeler

Taxonomy & Annotations:
github.com/oldvis/image-taxonomy

Paper:
arxiv.org/abs/2505.01724

OldVis Project:
github.com/oldvis